

dmap() manual

Activation of the function

If you have all the stuff on GitHub in a particular folder and dmap.R is also there, then it will be able to access the files it needs, which are specified in the beginning of the script. Activate the dmap() function and its required objects by writing `source("dmap.R")` after having set the working directory to the relevant folder.

Parameters

k: the number clusters desired, for instance $k = 4$ (default $k = 5$)

tree: the clusterization method; options are WardD, WardD2, complete, UPGMA, WPGMA (default UPGMA); note that quotation marks are not used, because the trees are objects

lonlim, latlim: specify a subarea setting these (default is the entire area covered by DARJa); for instance `lonlim=c(27.8,31.5)`, `latlim=c(58,60)` shows the diversity in the NW and `lonlim=c(40,45)`, `latlim=c(58,60)` shows the Chuxloma “island”

AREA: choose one among 9 cells in a 3 x 3 to zoom in on; options: "SW", "CW", "SW", "NC", "CC", "SC", "NE", "CE", "SE"

lofvals: if set to TRUE, lof scores will be employed as a way to identify outliers; lof scores are scores of how much a location diverges geographically from its fellows cluster members (=local outlier factor); **minpnts** is the number of nearest neighbors used in defining the local neighborhood of a point (includes the point itself); it needs to be above 63 to recognize the Chuxloma outliers; the **lofcutoff** could be lowered to include more of the points in the north, but then it will also include some dots that just happen to be a bit removed from fellow cluster members for geographical reasons

SVM: if set to TRUE, a support vector machine will be employed as a way to identify outliers; they are identified as such if they are not predicted by the model; doesn't work so well, perhaps might work better with some parameter tuning

DBSCAN: if set to TRUE; this identifies noisy points, which can be considered outliers; results are rather similar to lof

KNN: if set to TRUE, this counts the number of neighbors belonging to the same group; if this is below the SGC (same group count) threshold the location is counted as an outlier; didn't immediately manage to tune parameters to get a meaningful result

GMM: if set to TRUE, a Gaussian Mixture Models will produce a log-likelihood for the classification of a given point in its particular cluster; if this log-likelihood falls beneath a certain max probability (MP) cutoff it is defined as an outlier; somehow doesn't work well

LINES: if set to TRUE, the area is separated into a 3 x 3 grid by lines; only works when `GEOPLOT=FALSE`

GEOPLOT: if set to TRUE, display a ggplot2 map; if set to FALSE, displays a map without geographical projection, lon and lat being treated as plain Cartesian coordinates; in the GEOPLOT=FALSE mode, labels showing IDs are added automatically when looking for outliers and zooming in on a certain area

FILL: color of land

SCALE: "L" for high resolution (showing more rivers); "M" for medium resolution (fewer rivers); "S" for low resolution (nearly no rivers)

GENERICPLOT: if set to TRUE just plot the locations of DARJa; when exporting the image a good width to choose is 800

MARGIN: adjusts the margin for the purpose of GENERICPLOT

POINTSIZ: adjusts the point size for the purpose of GENERICPLOT

CITYSIZE: adjusts the size of symbol for anchoring cities for the purpose of GENERICPLOT

CITYSHAPE: adjusts the symbol for anchoring cities for the purpose of GENERICPLOT

BIOMEPL: if set to TRUE, display the biomes of the DARJa area, also outputting a legend to the console

FT: if this is some feature (e.g., "M_6"), a map will be created showing the distribution of values of that feature; note that multiple values cannot be shown, when they occur, some random value will overwrite the others; if FT is some feature value (e.g., "M_6_1"), the distribution of this value will be shown

Detail on outliers

When one of the outlier identifying functions is turned on the following information will be added to the dataframe with outgroup information: outgroup, id_inmate, orig_id_inmate, id_outmate, orig_id_outmate, hamming_inmate, hamming_outmate. A text file called outliers.txt will contain all information.

Detail on anchoring cities in the generic plot

A data frame called cities used for plotting some major cities in the area was prepared in the following way, which could be modified to make change to the cities chosen for display:

```
library(toponym) - see https://github.com/Lennart05/toponym
```

```
top(strings=c("Saint Petersburg", "Vologda", "Pskov", "Moscow",  
"Nizhniy Novgorod", "Smolensk", "Tambov", "Belgorod", "Chukhloma",  
"Velikiy Novgorod", "Velikiy Ustyug"), countries="RU",  
name="cities_all")
```

that produces a data frame cities_all from geonames.org, but with some cities whose names are synonymous, so the right ones are filtered by their geonames IDs

```
ids <- c(498817, 472459, 504341, 524901, 520555, 491687, 484646,  
578072, 567695, 519336, 476062)  
w_ids <- match(ids, cities_all$geonameid)  
cities <- cities_all[w_ids,c("name", "latitude", "longitude",  
"group")]
```

There is a column called group, which is reused here with the same name but just given a factor of 1 as values

```
cities$group <- as.factor(1)  
save(cities, file="cities.RData")
```